

Experiment No. 2

Experiment Name - Exploratory Data Analysis (EDA) on a Real-World Business Dataset using Python

Steps

NAME : Atul Krishna

COURSE : B.Tech CSE

SECTION - B

SUBJECT : Mastery in Data Analytics and Visualizations and Career Advancement

ROLL NO. : 14

CU ID : CU24250231

1. Select a real-world dataset such as the Netflix Titles Dataset, Superstore Sales Dataset, or IBM HR Employee Attrition Dataset.

For this experiment, we use the Superstore Sales Dataset.

The dataset contains business-related information such as:

- Sales
- Profit
- Quantity
- Category
- Sub-Category
- Region
- Customer information
- Product information

2. Import the required Python libraries (Pandas, NumPy, Matplotlib, and Seaborn) and load the cleaned dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
uploaded = files.upload() # select your Titanic CSV when prompted

import os
print(os.listdir()) # confirm the exact filename

print("Dataset loaded successfully!")
```

Choose Files Superstore.csv
Superstore.csv(text/csv) - 2287806 bytes, last modified: 8/31/2026 - 100% done
Saving Superstore.csv to Superstore (1).csv
['.config', 'Superstore.csv', 'Superstore (1).csv', 'sample_data']
Dataset loaded successfully!

```
df = pd.read_csv("Superstore.csv", encoding="latin1")

print("Dataset loaded successfully!")
print("Shape of dataset:", df.shape)
```

Dataset loaded successfully!
Shape of dataset: (9994, 21)

3. Explore the dataset by examining its dimensions, data types, column names, and summary statistics using functions such as info(), describe(), and value_counts().

```
print("Column Names:")
print(df.columns)

print("\nDataset Information:")
df.info()
```

```
print("\nSummary Statistics:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Column Names:
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
       'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
       'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
       'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],
      dtype='object')
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row ID                9994 non-null  int64
1   Order ID              9994 non-null  object
2   Order Date            9994 non-null  object
3   Ship Date             9994 non-null  object
4   Ship Mode             9994 non-null  object
5   Customer ID           9994 non-null  object
6   Customer Name         9994 non-null  object
7   Segment               9994 non-null  object
8   Country               9994 non-null  object
9   City                  9994 non-null  object
10  State                 9994 non-null  object
11  Postal Code           9994 non-null  int64
12  Region                9994 non-null  object
13  Product ID            9994 non-null  object
14  Category              9994 non-null  object
15  Sub-Category          9994 non-null  object
16  Product Name          9994 non-null  object
17  Sales                 9994 non-null  float64
18  Quantity              9994 non-null  int64
19  Discount              9994 non-null  float64
20  Profit                9994 non-null  float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

```
Summary Statistics:
      Row ID  Postal Code  Sales  Quantity  Discount \
count  9994.000000  9994.000000  9994.000000  9994.000000  9994.000000
mean    4997.500000  55190.379428   229.858001    3.789574    0.156203
std     2885.163629   32063.693350    623.245101    2.225110    0.206452
min         1.000000    1040.000000     0.444000    1.000000    0.000000
25%     2499.250000   23223.000000    17.280000    2.000000    0.000000
50%     4997.500000   56430.500000    54.490000    3.000000    0.200000
75%     7495.750000   90008.000000    209.940000    5.000000    0.200000
max     9994.000000   99301.000000   22638.480000   14.000000    0.800000

      Profit
count  9994.000000
mean    28.656896
std     234.260108
min    -6599.978000
25%      1.728750
50%      8.666500
75%     29.364000
max     8399.976000
```

4. Analyze the distribution of numerical features using histograms and density plots to understand central tendency and data spread.

```
numerical_columns = df.select_dtypes(include=np.number).columns

df[numerical_columns].hist(
    figsize=(5, 4),
    bins=20
)

plt.tight_layout()
plt.show()
```



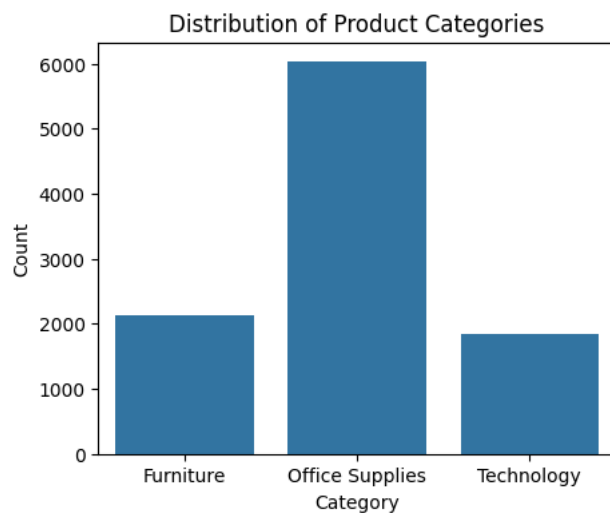
5. Visualize categorical variables using bar charts and count plots to identify the most frequent categories.

```
plt.figure(figsize=(5, 4))

sns.countplot(
    data=df,
    x="Category"
)

plt.title("Distribution of Product Categories")
plt.xlabel("Category")
plt.ylabel("Count")

plt.show()
```



6. Generate a correlation matrix and heatmap to identify relationships between numerical variables and detect highly correlated features.

```
numeric_df = df.select_dtypes(include=np.number)

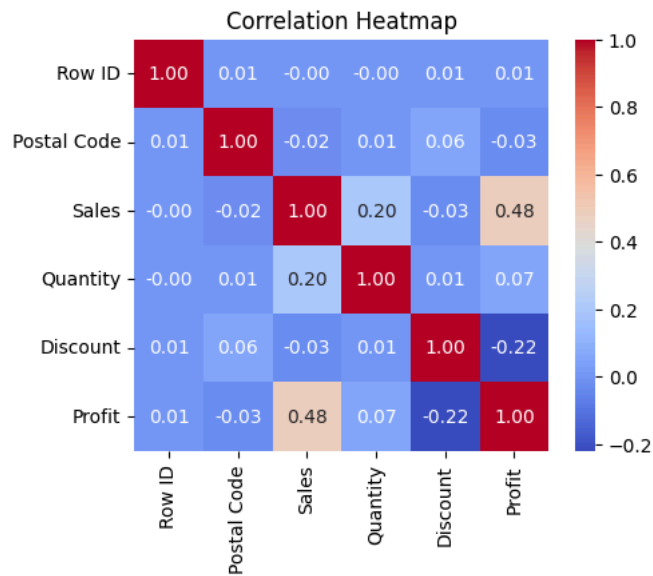
correlation = numeric_df.corr()

plt.figure(figsize=(5, 4))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")

plt.show()
```



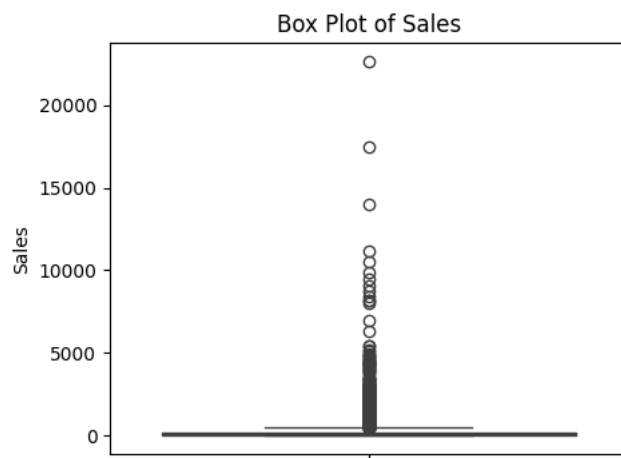
7. Create box plots and scatter plots to identify outliers and study relationships between different variables.

```
plt.figure(figsize=(5, 4))

sns.boxplot(
    data=df,
    y="Sales"
)

plt.title("Box Plot of Sales")
plt.ylabel("Sales")

plt.show()
```

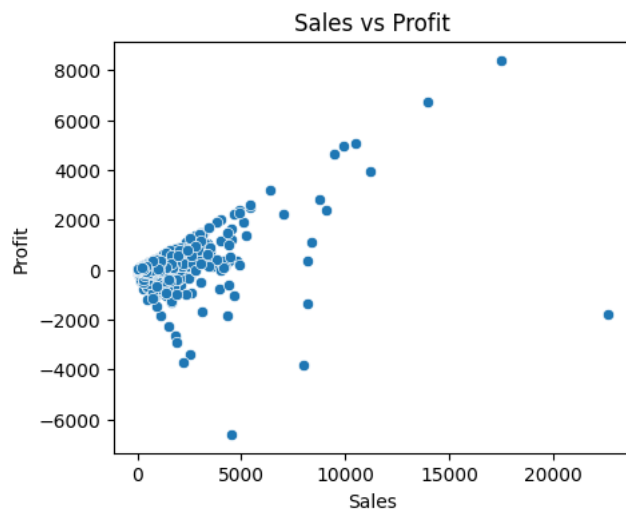


```
plt.figure(figsize=(5, 4))

sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit"
)

plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")

plt.show()
```



▼ 8. Perform univariate, bivariate, and multivariate analysis to uncover trends and business patterns within the dataset.

```
print("Sales Statistics:")
print(df["Sales"].describe())

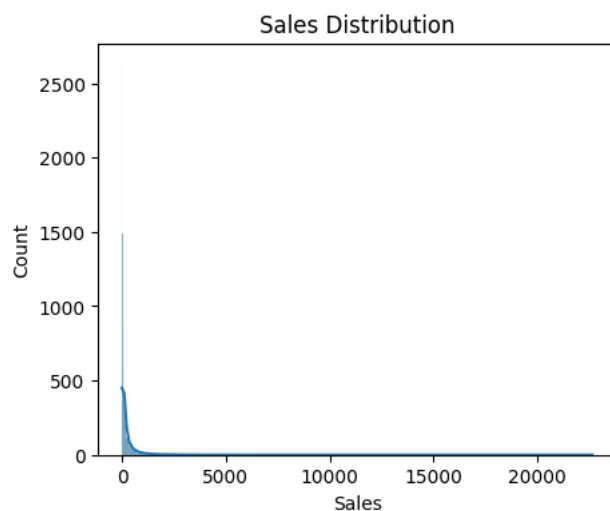
plt.figure(figsize=(5, 4))

sns.histplot(
    data=df,
    x="Sales",
    kde=True
)

plt.title("Sales Distribution")

plt.show()
```

```
Sales Statistics:
count      9994.000000
mean        229.858001
std         623.245101
min          0.444000
25%         17.280000
50%         54.490000
75%        209.940000
max       22638.480000
Name: Sales, dtype: float64
```

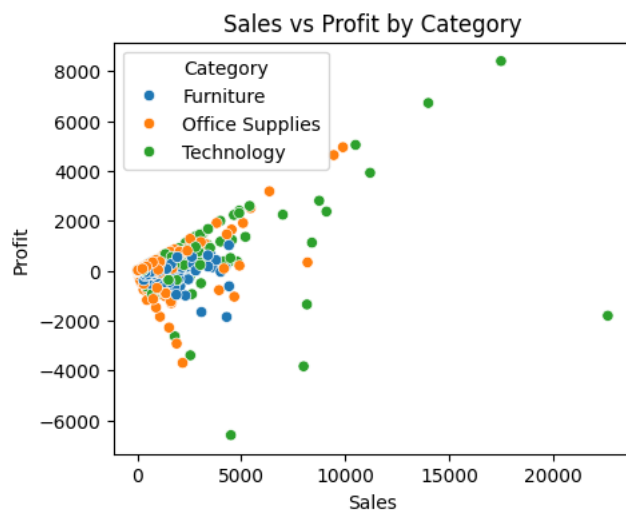


```
plt.figure(figsize=(5, 4))

sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
    hue="Category"
)

plt.title("Sales vs Profit by Category")
```

```
plt.show()
```

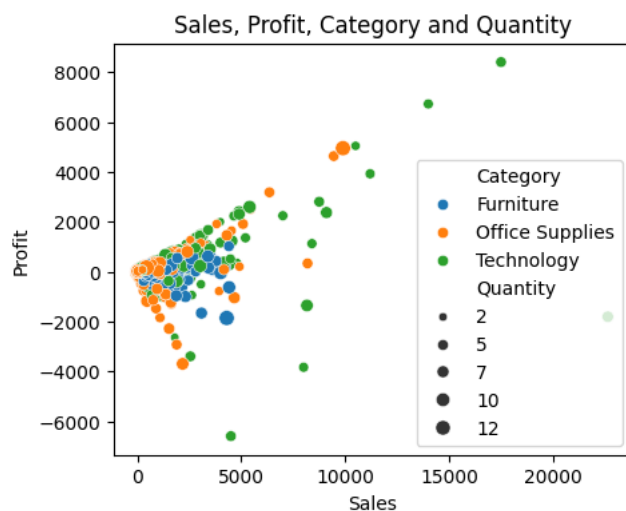


```
plt.figure(figsize=(5, 4))

sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit",
    hue="Category",
    size="Quantity"
)

plt.title("Sales, Profit, Category and Quantity")

plt.show()
```



✓ **9. Summarize key observations and identify factors that significantly influence the target variable or business objective.**

```
print("Average Sales:", df["Sales"].mean())
print("Average Profit:", df["Profit"].mean())
print("Maximum Sales:", df["Sales"].max())
print("Maximum Profit:", df["Profit"].max())

print("\nMost Common Category:")
print(df["Category"].mode()[0])

print("\nMost Common Region:")
print(df["Region"].mode()[0])
```

```
Average Sales: 229.85800083049833
Average Profit: 28.65689630778467
Maximum Sales: 22638.48
Maximum Profit: 8399.976
```

```
Most Common Category:
Office Supplies
```

Most Common Region:
West

Key Observations

From the exploratory analysis, the following information can be obtained:

1. The average sales value represents the typical sales generated per transaction.
2. Average profit indicates the average profitability of transactions.
3. Maximum Sales identifies the highest-value transaction present in the dataset.
4. Maximum Profit identifies the most profitable transaction.
5. The most common category identifies the category with the greatest number of records.
6. The most common region shows which geographical region appears most frequently.
7. The scatter plot indicates that higher sales do not always guarantee higher profit.
8. The box plot reveals the presence of unusually high sales transactions.
9. The correlation heatmap identifies relationships between numerical business variables.
10. Category-wise visualization helps compare product performance.

10. Prepare a concise report highlighting important findings and recommendations based on the exploratory analysis.

Findings

EDA was successfully performed on the Superstore Sales Dataset.

The analysis showed the distribution of sales and other numerical variables, frequency of different product categories, relationships between numerical features, presence of potential outliers, and the relationship between Sales and Profit.

Univariate, bivariate, and multivariate analyses were performed to better understand different business patterns.

Business Recommendations

Based on the exploratory analysis:

- Product categories generating high profits should receive greater attention.
- High-sales but low-profit transactions should be investigated.
- Loss-generating transactions should be analyzed to determine whether discounts or other factors are responsible.
- High-performing regions and categories can be prioritized for business expansion.
- Unusual sales values should be examined before predictive modeling.
- Sales, Profit, Discount, Quantity and Category can be studied further for improving business decision-making.

Questions and Answers

Q1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?

Answer - Exploratory Data Analysis or EDA is the process of examining and analyzing a dataset to understand its structure, distributions, relationships, patterns, and anomalies.

EDA is performed before machine learning because it helps identify:

- Missing values
- Outliers
- Skewed distributions
- Relationships between variables
- Incorrect data types
- Data quality problems

It allows analysts to properly understand and prepare the dataset before building predictive models.

Q2. Differentiate between univariate, bivariate, and multivariate analysis with suitable examples.

Answer -

Analysis	Meaning	Example
Univariate	Analysis of one variable	Distribution of Sales
Bivariate	Analysis of two variables	Sales vs Profit
Multivariate	Analysis of more than two variables	Sales, Profit, Category and Quantity

Univariate analysis studies only one variable.

Example: Sales Distribution

Bivariate analysis studies the relationship between two variables.

Example: Sales vs Profit

Multivariate analysis studies several variables simultaneously.

Example: Sales + Profit + Category + Quantity

Q3. What insights can be obtained from a correlation heatmap?

Answer - A correlation heatmap shows the strength and direction of relationships between numerical variables.

It helps identify:

- Positive relationships
- Negative relationships
- Weak correlations
- Strong correlations
- Highly correlated features
- Potentially redundant variables

For example, it can help determine whether Sales has a positive or negative relationship with Profit, Quantity or Discount.

Q4. Explain the purpose of histograms, box plots, and scatter plots in EDA.

Answer -

Histogram - A histogram is used to study the distribution and frequency of numerical data.

It helps identify:

- Distribution
- Central tendency
- Spread
- Skewness

Box Plot -

A box plot displays:

- Median
- Quartiles
- Spread
- Potential outliers

It is particularly useful for identifying unusually high or low values.

Scatter Plot - A scatter plot is used to analyze the relationship between two numerical variables.

For example: Sales vs Profit

can show whether Profit generally increases with Sales.

Q5. How can EDA help identify data quality issues before analysis?

Answer - EDA can identify several data quality problems, including:

- Missing values
- Duplicate records
- Outliers
- Incorrect data types
- Unusual values
- Inconsistent categories
- Skewed distributions
- Invalid values

By identifying these problems before analysis, the dataset can be properly cleaned and prepared for further processing and predictive modeling.

Q6. Why is correlation important in predictive analytics? Can correlation imply causation?

Answer - Correlation is important because it measures the relationship between numerical variables.

It can help identify:

- Variables useful for prediction
- Strongly related features
- Redundant variables
- Potential predictors

However, correlation does not imply causation.

Two variables may be correlated because of another hidden variable or coincidence. Therefore, correlation alone cannot prove that one variable causes another.

Q7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.

Answer - For categorical variables, suitable visualizations include:

- Bar charts
- Count plots

For example, a count plot can show how many records belong to each Product Category.

For numerical variables, suitable visualizations include:

- Histograms
- Box plots
- Scatter plots

A histogram helps understand numerical distributions, a box plot identifies outliers, and a scatter plot studies relationships between numerical variables.

Q8. What business insights can be derived from the Superstore dataset through EDA?

Answer - The Superstore dataset can provide business insights such as:

- Which product categories have the highest sales
- Which categories generate the most profit
- Which regions contain the greatest number of transactions
- Relationship between Sales and Profit
- Identification of loss-generating transactions
- Identification of unusually large sales
- Distribution of orders across categories
- Relationship between quantity and sales
- Potential effect of other numerical variables on profitability

These insights can help management make better data-driven business decisions.

Q9. How does EDA contribute to feature selection and model building?

Answer - EDA helps determine which variables may be useful for predictive modeling.

Correlation analysis can identify highly related variables, while distribution analysis helps identify skewed features and box plots can reveal outliers.

EDA therefore helps in:

- Selecting useful features
- Detecting redundant features
- Understanding relationships
- Identifying outliers
- Detecting skewness
- Preparing variables for machine learning
- Improving model quality

Q10. What challenges might arise while performing EDA on large-scale real-world datasets?

Answer - Some common challenges include:

- Very large dataset size
- High memory consumption
- High computational requirements
- Missing or inconsistent data
- Large numbers of features
- Difficulty visualizing millions of records
- Outliers and noisy observations
- Complex relationships among variables
- Long processing times
- Difficulty interpreting large numbers of visualizations

Therefore, large datasets may require efficient data-processing techniques, sampling methods, scalable tools, and appropriate visualization strategies.